

RES-5 dressed three-loop skeleton: the dressed loop parameter is $O(1)$

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The 2PI-skeleton framing (res5-higherloop-commonmode-bound) reduced RES-5 to the skeleton domination $s_\ell \leq s < \frac{1}{2}$ with the favourable base $s_3 \sim 4\%$ (sunset). The operator directed the next term: bound the dressed three-loop skeleton ratio $s_4 = |\Delta F_{3\text{-loop}}^{\text{pd}}|/\Delta F_{\text{margin}}$, and ideally show $s_4 \ll s_3$. This note evaluates the skeleton RATIO via the robust dressed loop parameter and reports the honest verdict: the ratio is $O(1)$ (marginal), so $s_4 \sim s_3$ and the geometric domination does not hold – RES-5 needs resummation. Scope: the dressed loop parameter; anchor μ^2 , endpoint.

2. The dressed loop parameter is $O(1)$ (marginal)

The skeleton ratio $s_{\ell+1}/s_\ell$ is the dressed per-loop factor, $\sim \lambda' \times (\text{dressed bubble})$. At the operating endpoint ($\hat{r} = 0.337$, $\lambda' = 8.06$):

$$\lambda' M_d = \lambda' \int G_d = 0.78, \quad \lambda' B_d = \lambda' \int G_d^2 = 1.03 = O(1).$$

The dressed loop parameter is ≈ 1 – MARGINAL, at the convergence boundary. This is the Brazovskii strong-fluctuation regime: even with the self-consistent dressing, the loop expansion is not controlled by a small parameter.

3. Consequence: $s_4 \sim s_3$, the series does not decay

$s_4 \sim s_3 \cdot (\lambda' B_d) \sim 0.04 \times 1.03 \approx 0.041 \sim s_3 \quad (\text{NOT} \ll s_3).$

Each skeleton term stays small ($\sim 4\%$, from the a_0 / common-mode suppression of the DIFFERENCE), so the per-term condition $s_\ell < \frac{1}{2}$ plausibly holds at every order. But the RATIO is ~ 1 , so the terms do NOT geometrically decay: the geometric sum $\sum_{\ell \geq 3} s_\ell$ with ratio ~ 1 is MARGINAL (it does not converge cleanly). The favourable base s_3 does not, by itself, bound the tail.

4. RES-5 needs resummation (the honest verdict)

A dressed loop parameter $O(1)$ means the matched-order-to-exact remainder is a MARGINAL strong-coupling series. Its bound cannot come from a geometric skeleton-by-skeleton estimate; it requires a RESUMMATION – the 2PI self-consistent effective action carried beyond the sunset,

or a Borel / optimal- truncation treatment of the marginal series. The per-term a_0 -smallness (the common-mode cancellation) is the reason the problem is not hopeless, but the $O(1)$ ratio is the genuine strong-coupling obstruction. **B1 closure** $\iff$ a controlled resummation of this marginal dressed-skeleton series.

5. Status

T3 PROOF SKETCH. The dressed loop parameter $\lambda' B_d = 1.03 = O(1)$ is pinned (robust); $s_4 \sim s_3$ (ratio ~ 1 , not decaying); per-term $s_\ell < \frac{1}{2}$ but the geometric sum is marginal. **OPEN (the genuine frontier):** a 2PI / Borel RESUMMATION of the marginal dressed-skeleton series – a dedicated strong-coupling QFT computation, not an elementary geometric bound. No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “ $s_4 \sim s_3$ is a scaling estimate, not the actual three-loop integral.” VALID: the dressed loop parameter $\lambda' B_d = 1.03$ is robust (a converged integral), and gives the RATIO; the precise s_4 needs the actual three-loop skeleton integral, but its SCALE is fixed by the loop parameter, so the marginal-ratio conclusion holds at the scaling level.
- (β) “Marginal ratio means RES-5 fails.” DISMISSED: marginal (≈ 1) is not > 1 ; with the a_0 -suppressed per-term smallness, the series may still sum (optimal truncation / Borel), and the selection has survived at every computed order (SC-SCOPE). It means a geometric bound is the wrong tool – resummation is needed – not that RES-5 is false.
- (γ) “Does this change B1?” DISMISSED: it characterises the residual (marginal, needs resummation), not any tier. B1 stays T6 on {H-LAYER}.

Quantitative sanity check (mandatory): *magnitude* – $\lambda' B_d = 1.03 = O(1)$ (marginal); *ratio* – $s_4/s_3 \sim \lambda' B_d \sim 1$ (not decaying); *per-term* – $s_4 \sim 0.04 < \frac{1}{2}$ (a_0 -suppressed); *regime* – Brazovskii strong fluctuations give an $O(1)$ dressed loop parameter, as expected for a fluctuation-driven first-order transition.

7. Result footer

Result ID:	B1-RH-ENUM / RES-5 dressed three-loop skeleton (loop param)
Precise statement:	The dressed skeleton ratio $s_{\{1+1\}}/s_1 \sim \lambda' B_d = \lambda' \int G_d^2 = 1.03 = O(1)$ (marginal). Hence $s_4 \sim s_3 * 1.03 \sim s_3$ (NOT $\ll s_3$): each term $\sim 4\%$ (a_0 -suppressed) but the series ratio is ~ 1 , so it does not geometrically decay. Per-term $s_1 < 1/2$ plausibly holds; the geometric SUM is marginal. RES-5 closure needs a 2PI/Borel RESUMMATION of the marginal dressed-skeleton series, not a geometric bound.
Scope:	Dressed loop parameter (robust) + scaling estimate of s_4 ; anchor μ^2 , endpoint. The precise three-loop integral and the resummation are OPEN.
Dependencies:	res5-higherloop-commonmode-bound (the 2PI framing); SC-SCOPE (s_3 sunset); Math424-AddA (rR, λ' , G_d).
Evidence grade:	ANALYTIC + EXECUTED (res5_dressed_loop_parameter.py 4/4).
Reproduction command:	python codes/vacuum/res5_dressed_loop_parameter.py

Expected output: $\text{lam}' M_d=0.78, \text{lam}' B_d=1.03 \ (0(1)); s_4 \sim s_3 \sim 0.04; 4/4.$

Falsification gate: A dressed loop parameter $\ll 1$ (would restore the geometric bound) or $\gg 1$ (would signal a genuine divergence); a resummation giving $|\text{sum}_{\{l \geq 3\}}| \geq \Delta F_{\text{margin}}$ (would threaten the selection at exact order).

Tier before / after: No tier flip. B1 T6 on {H-LAYER}. RES-5: the dressed skeleton series is MARGINAL (loop param $0(1)$); closure needs resummation.

No-overclaim statement: This does NOT close RES-5 and does NOT change B1's tier. It establishes that the dressed-skeleton series is marginal (loop parameter $0(1)$) and that geometric bounds are the wrong tool; a 2PI/Borel resummation is the genuine residual. The per-term a_0 -smallness is real; the $0(1)$ ratio is the strong-coupling frontier.

Next required action: A dedicated 2PI self-consistent (or Borel / optimal-truncation) RESUMMATION of the dressed-skeleton DIFFERENCE series; this is a major strong-coupling QFT effort, the genuine H-LAYER/RES-5 closure frontier.